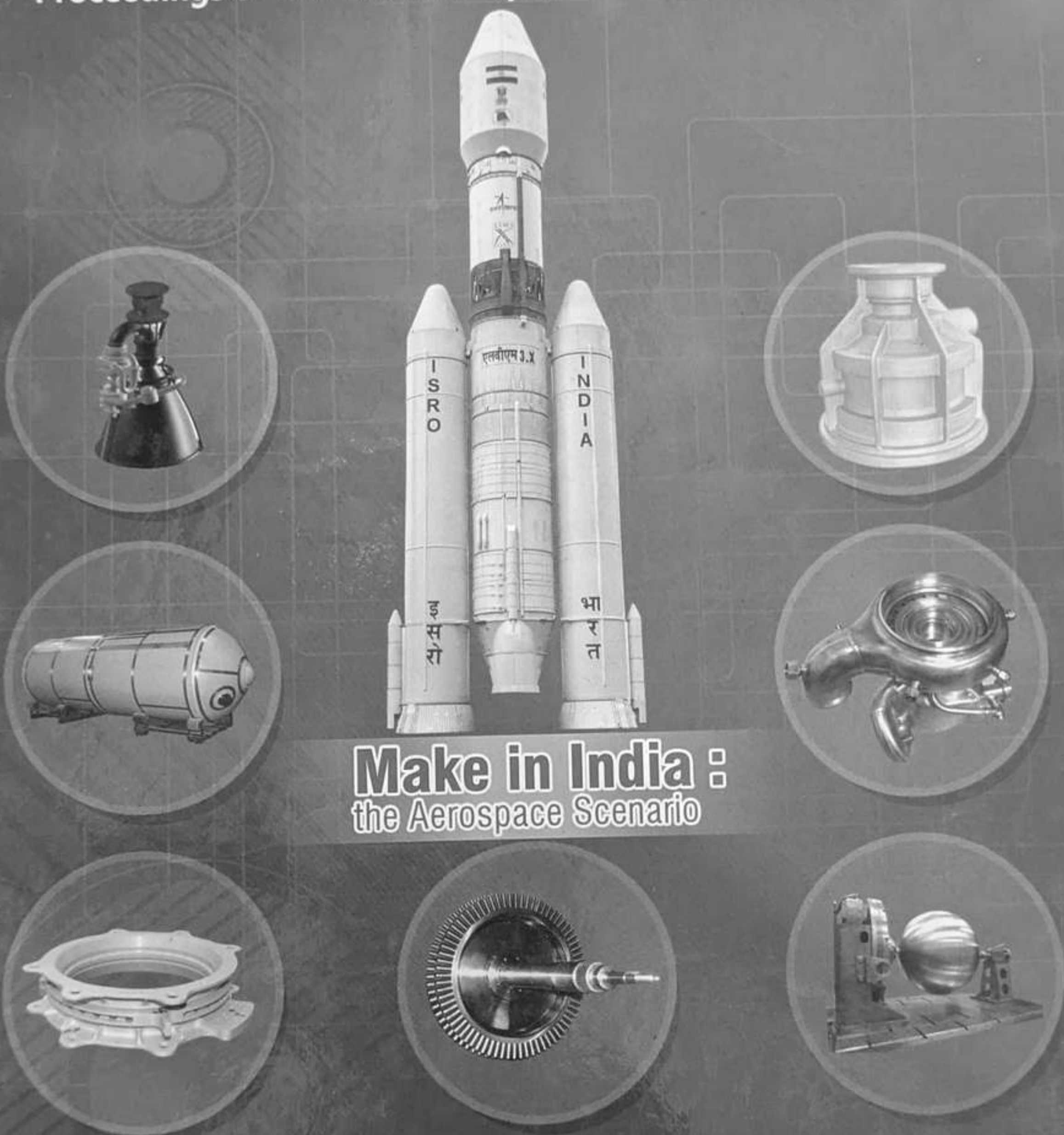


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A TYPICAL DEVELOPMENT OF HIGH AMPERAGE SLIP RING

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ABSTRACT

Godrej & Boyce Manufacturing Limited had approached us to develop a high amperage slip ring for commercial applications with specifications of 300A & 5V DC [1]. Legend Technologies Pvt Ltd has been involved in Slip Ring Design, Development & Testing for almost 25 years, but this was a unique challenge to us. We had a transfer of technology from ISRO about the basic design of slip rings [2].

Introduction

The primary challenges faced by us were Silver Graphite powder formation after certain period of usage & it leads to short-circuit between the channels. Continuity problems faced between rotor & stator. Water ingress inside the slip ring. Frequent maintenance & Lubrication. Slippage of brush blocks between the channels. More power required because of friction. Contact between stator and rotor is being done using Beryllium copper wire and Electrode grade copper rings with U groove technology. Insulation between stator and rotor is done using special Epoxy potting compound with fixtures. Gold plating on Contacts [Rings & Brush wires]. Printed Circuit Board [specially designed for application]. IP [Ingress protection] sealing.

Electrode grade copper and beryllium copper wire used as contacts (B/w rotor and stator) makes no powder formation & better conductivity. Gold plating to ensure high conductivity & long life. Use of PC board to ensure continuity & better assembly features.



Ingress protection using sealants and O-rings to protect the slip ring from dust & water.

Wear and tear of contacts is avoided. Continuity is ensured at all times. Potting compounds provides better insulation. IP rating makes the slip ring operate for longer time in all environment. Life of slip ring is increased to 15 years. Maintenance free. Iterative design.

Design

The design of the slipring involved many of the critical parameters such as Bearings, Brush wires, Electrical, Lead Wires, Structural parameters & thermal.

Material Selection

Each of the materials for the different components of the slip ring have been carefully studied and picked for the correct application. There are 11 unique components that we developed for this application. The shaft was made into stainless steel to accommodate the rotating stressed involved. The insulation for the shafts were provided through means of Polyurethane compounds for easy adherence and curing. The sleeves and the skeleton; most of the non-loaded components are made with aluminium to keep the weight down. The brush wires are made of beryllium copper and were gold plated to prevent the formation of dust, better conductivity & longer life. All the components have a reason and a use for their choice

Preliminary Design

The preliminary design was primarily done through excel sheets for their ease of manipulation. Various factors were considered before moving to the final design. A sample calculation sheet is attached for the better understanding of the thermal characteristics of the product.



Table 1: A Preliminary Calculation Sheet

Volt	Vdc	5	
Length of Wire	L	0.1	m
Loss%		5%	
Area	A	0.5	mm ²
Ampere	I	30	A
Time	T	30	hr
Pi	PI	3.141593	
Density Cu		8960	Kg/m ³
Specific Heat Cu	Cp	390	J/Kg K
Modulus of Elasticity Be-Cu	E	1.38E+11	N/m ²
Static Friction (Au-Au)	Mu-s	0.3	
Speed	N	200	rpm
Radius of Contact	r	34	mm
Resitivity	Ro	1.68E-08	ohm-m
Diameter	d	0.797885	mm
Resistance	R	3.36E-09	ohm
Power Loss	W	3.02E-06	W(J/s)
Power	W-hr	0.326592	J
Volume	V	5E-08	m ³
Mass	m	0.000448	Kg
Temp Rise	dT	1.869231	K

DETAIL DESIGN

Each component was designed for function, manufacturing and assembly. The assembly sequence was also pre-planned to allow for proper assembly. The below given assembly is the final detail model for which manufacturing drawings were released.

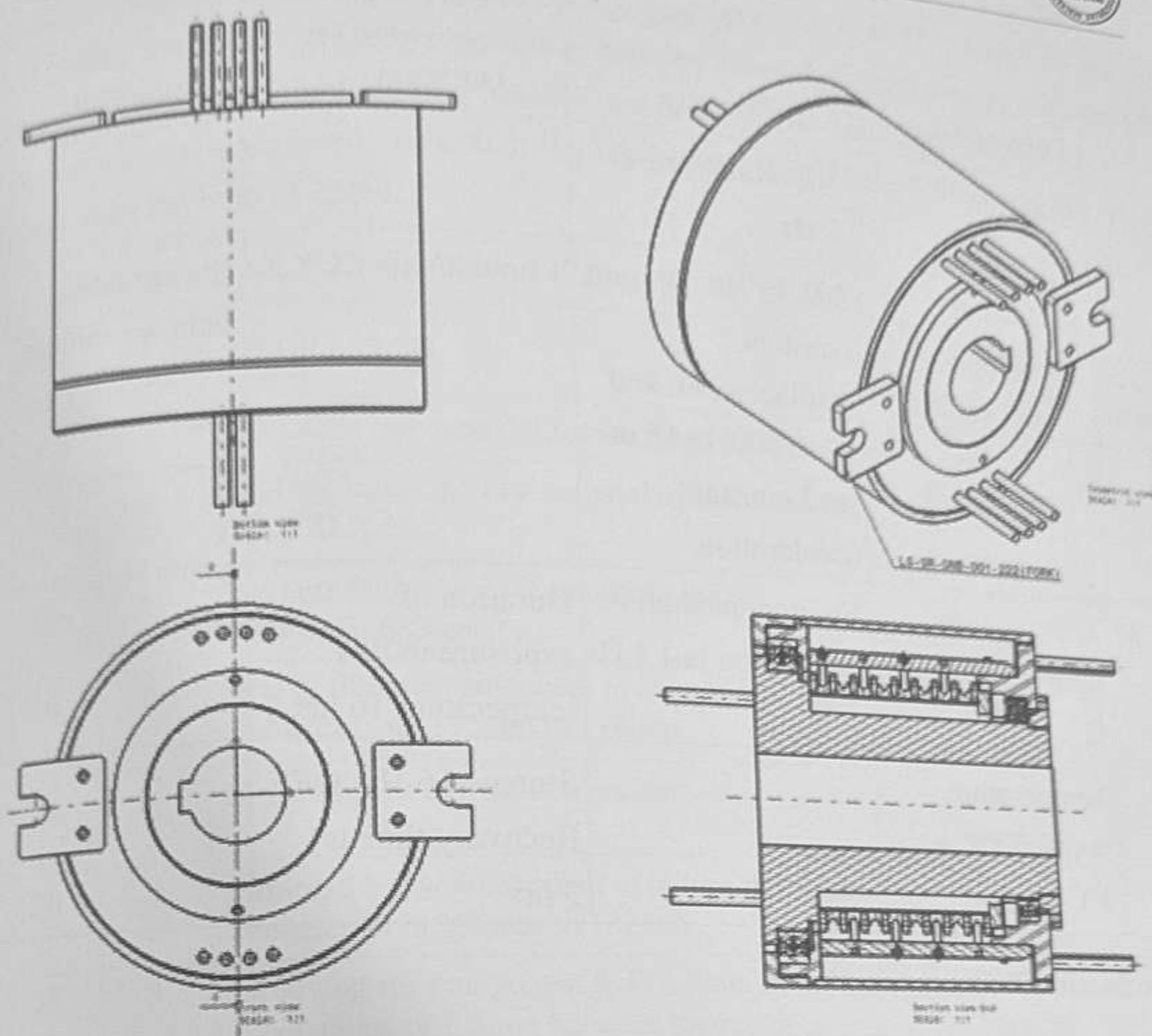


Figure 1: Assembly of the Slipring [3]

Analysis

The vibrational testing was a concern, but after we were able to get the natural frequency and the different modes of it, we were able to check if any resonant frequencies were present and cross referenced it to check if any would lead to resonant failure.



Table 2: Analysis Load Cases

	Type of Test	Severity	Duration	Comments
1	200 rpm	Vibrations are @ 3 Hz		
2	Sinusoidal Vibration	5 to 13 Hz, ± 6 mm constant displacement, and 13 to 500Hz, 15 m/sec ² constant acceleration	1 hours /axis (X,Y,Z)	Sweep rate: 1 octave / minute
3	Temperature range: 55°C + 3°C	Equipment shall be ON during last 1 Hr	Duration of exposure to this temperature 16 hrs	
4	Temperature range :70°C + 3°C		Storage 16 Hrs OFF Recovery time is 3 hrs	

Production

The fabrication of the prototype involved different steps such as Turning, Machining, Electro-discharge machining, broaching, and drilling. One typical example of fabrication that can be overlooked is the necessary for pre-planning. The shaft had perpendicularity and concentricity for the proper seating of the bearings and the contact rings. In the actual VMC that was used, if the setting is changed for each step of machining for instance once the roughing operation is completed and the component is removed and even after correctly setting the centering of the shaft will definitely vary according to the chuck movement. All the dimensions required have to be machined in one setting only to get the perpendicularity and concentricity.

Likewise, some of the typical processes are given below for reference. The special processes can be delved into in this paper. Let us examine the EDM necessary for machining the electrode grade copper. Since copper has a tensile strength much lower



when compared to all the other usually machined metals. It is very tough to get the necessary tolerances, therefore we had to turn the ID of the contact rings with the full shaft and then with the help of a mandrel we turn the OD and the necessary grooving operations are completed. To match the final dimensions we cut the rings out of the shaft with the help of EDM.

Table 3: Process Plan Sheet [3]

OPN. NO	STAGES	OPERATION	WORK CENTER	REMARK
5		Inspect the raw material for the given specifications.	QC	
10	1	Load the inspected raw material of size Ø115ID X Ø135OD X 69.5	CNC Turning	
15	2	One side facing is done to reduce the thickness from 69.5mm to 68.5mm	CNC Turning	
20	3	turning the raw component to increase the inner diameter from 115mm to 118mm	CNC Turning	
25	3	Mandrel is fit to hold component.	CNC Turning	
30	4	turning the raw component to reduce the outer diameter from 135mm to 132mm	CNC Turning	
35	5	Grooving the component to Ø 4.5mm at eight places by distance of 8.5mm between them.	CNC Turning	
40	6	Other side facing is done to reduce the thickness from 68.5mm to 67.5mm	CNC Turning	
45	7	Electron discharge machining (EDM) carried to separate each rings.	Cutting	
50		Inspect the turning components as per drawing and record the data in inspection report	QC	

Testing and after sales services

The most vital part of our problems was in the testing, as even CPRI were unable to provide us the required 300A at 5VDC. As the quality assurance plan indicates the different electrical parameters that had to be tested for the product to be certified.



Table 4: Quality Assurance Plan [4]

Operation	Parameters	Method	Extend of Check	Acceptance Standard
Inspection & Testing	Static Contact Resistance	Test Circuit & Oscilloscope	Each Channel will be checked	<100 milli Ohms per ring
	Dynamic Contact Resistance	Test Circuit & Oscilloscope	Each Channel will be checked	<150 milli Ohms per ring for 200 rpm
	Insulation Resistance Test	Megger	B/w any two rings B/w ring and body	> 200 Mega Ohms with 500V DC
	Continuity Test (Static)	Multi-meter	Continuity from Input to Output End	Verification of Continuity
	Continuity Test (Dynamic)	Multi-meter	Continuity from Input to Output End	Verification of Continuity
	Electrical Noise Test	Test Circuit, Power Supply, Variable Speed Drive Set, Oscilloscope	Unit to be rotated at 200rpm.	Noise should be <5mV/Amp/Ring
	Plating Test	XRF Method	All Contact Rings and the Brush Wires	>2 microns
	Performance Testing for One Hour at Rated Load		100%	External Lab Report
	Non-Destructive Test	Ultrasonic Examination (Pulse Echo Technique)	100%	NDT Report

Conclusion

This paper is presented to give an insight of how exactly a typical high amperage slipring is developed. Though the paper might not be a complete description of the activities involved in developing a product, it might be used as a blueprint to others who look for the processes, the thoughts, the actions that go into developing a functional product.



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